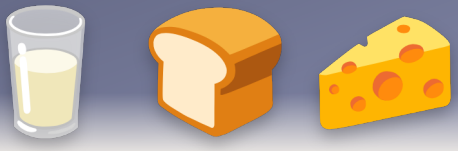


Tiny Helpers

II PUC · Ch 8 · Microbes in Human Welfare

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



Breakfast Friends

At breakfast, Prof. BAIO pointed at the curd. "**Lactobacillus** — lactic acid bacteria — turned milk into curd and added **vitamin B_{12}**!" Baker's yeast, **Saccharomyces cerevisiae**, made the bread fluffy. The big holes in Swiss cheese were made by **Propionibacterium sharmanii**.

💡 **REMEMBER**

Dosa and idli batter is also fermented by bacteria.



Factory Microbes

In factories, brewer's yeast made wine, beer and whisky. Alexander **Fleming** noticed a mould, **Penicillium notatum**, killing bacteria — the first **antibiotic**, penicillin. Chain and Florey made it into a medicine.

> 💡 **REMEMBER**
> **Fleming, Chain and Florey shared the Nobel Prize in 1945.**



Tiny Chemists

Microbes made acids: citric acid (Aspergillus niger), acetic acid (Acetobacter aceti), butyric acid (Clostridium butylicum) and lactic acid (Lactobacillus). **Streptokinase** dissolves blood clots, **cyclosporin A** helps organ-transplant patients, and **statins** lower cholesterol.

💡 **REMEMBER**

Lipases are used in detergents; pectinases and proteases clear fruit juices.



Cleaning the Water

At the sewage plant, **primary treatment** removed solids by filtering and settling. In **secondary treatment**, air was pumped in and microbes formed **flocs** that ate the organic matter, so the **BOD** fell. The settled **activated sludge** went to anaerobic digesters.

💡 **REMEMBER**

BOD — the higher it is, the more polluted the water.



Gas from Dung

In the village, cow dung went into a **biogas** plant. **Methanogens** like Methanobacterium broke it down without oxygen, making methane gas for cooking and lighting. These microbes also live in the rumen of cattle.

💡 **REMEMBER**

Biogas technology in India was developed by IARI and KVIC.



The Pest Police

Farmers called in **biocontrol** agents. Ladybirds ate aphids; dragonflies ate mosquitoes. **Bacillus thuringiensis** killed caterpillars. The fungus **Trichoderma** fought plant diseases, and **baculoviruses** attacked only specific insects.

💡 REMEMBER

Biocontrol is part of integrated pest management (IPM).

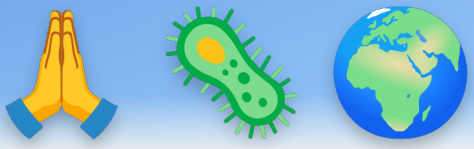


Natural Fertilisers

Biofertilisers fed the fields. **Rhizobium** lived in root nodules of legumes and fixed nitrogen. Azotobacter and Azospirillum fixed nitrogen in the soil. The fungus Glomus formed **mycorrhiza** to help roots take up phosphorus. Cyanobacteria like Anabaena and Nostoc enriched paddy fields.

💡 **REMEMBER**

Biofertilisers reduce the use of chemical fertilisers.



Thank You, Microbes!

As the sun set, Bindu wrote in her notebook: "Microbes give us curd, bread, medicines, clean water, fuel, pest control and healthy soil." Prof. BAIO nodded. "Not all germs are enemies — many are our tiny helpers."

💡 **REMEMBER**

Microbes are used in household products, industry, sewage treatment, biogas, biocontrol and biofertilisers.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

Lactobacillus p.1 · vitamin B_{12} p.1 · Saccharomyces cerevisiae p.1 · Propionibacterium sharmanii p.1 · Fleming p.2 · Penicillium notatum p.2 · antibiotic p.2 · Streptokinase p.3 · cyclosporin A p.3 · statins p.3 · primary treatment p.4 · secondary treatment p.4 · flocs p.4 · BOD p.4 · activated sludge p.4 · biogas p.5 · Methanogens p.5 · biocontrol p.6 · Bacillus thuringiensis p.6 · Trichoderma p.6 · baculoviruses p.6 · Biofertilisers p.7 · Rhizobium p.7 · mycorrhiza p.7



QUICK QUIZ

Do you remember?

1. Curd is formed by:

- Lactobacillus
- Penicillium
- Aspergillus
- Rhizobium

2. Swiss cheese holes are made by:

- Yeast
- Propionibacterium sharmanii
- Lactobacillus
- Trichoderma

3. Citric acid is produced by:

- Aspergillus niger
- Acetobacter aceti
- Clostridium
- Lactobacillus

4. Biogas is mainly:

- Carbon dioxide
- Methane
- Hydrogen sulphide
- Oxygen

5. A clot buster made by Streptococcus is:

- Statin
- Streptokinase
- Cyclosporin A
- Lipase

Answers: 1-Lactobacillus · 2-Propionibacterium sharmanii · 3-Aspergillus niger · 4-Methane · 5-Streptokinase